ENGINEERING PORTFOLIO

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B.S. Mechatronic Engineering
Minor in Computer Engineering

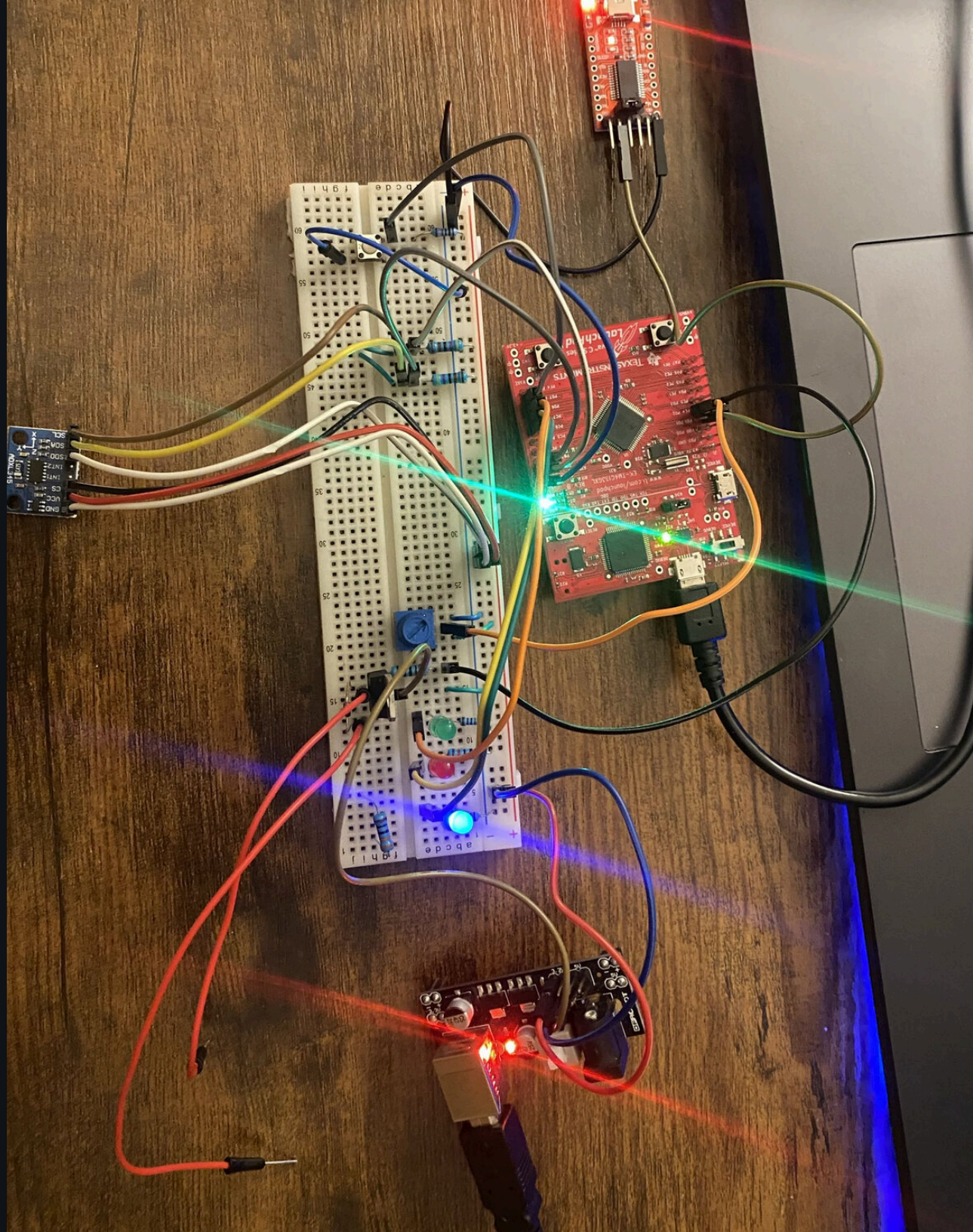
California State University, Chico · Spring 2027

CONTROLS
PI control · Pole placement · MATLAB / Simulink

EMBEDDED
Bare-metal C · ADC · PWM · Timers · Interrupts

HARDWARE & TOOLS
UART · I²C · LTspice · Oscilloscope

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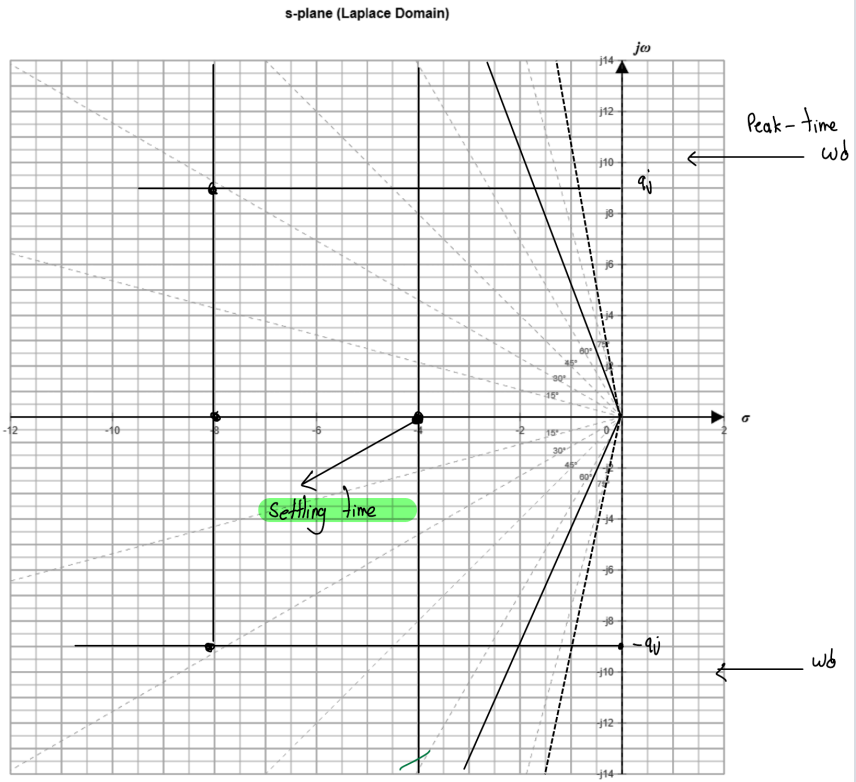
PI control of an RC plant: design

Pole placement · MATLAB PID Tuner · Op-amp implementation

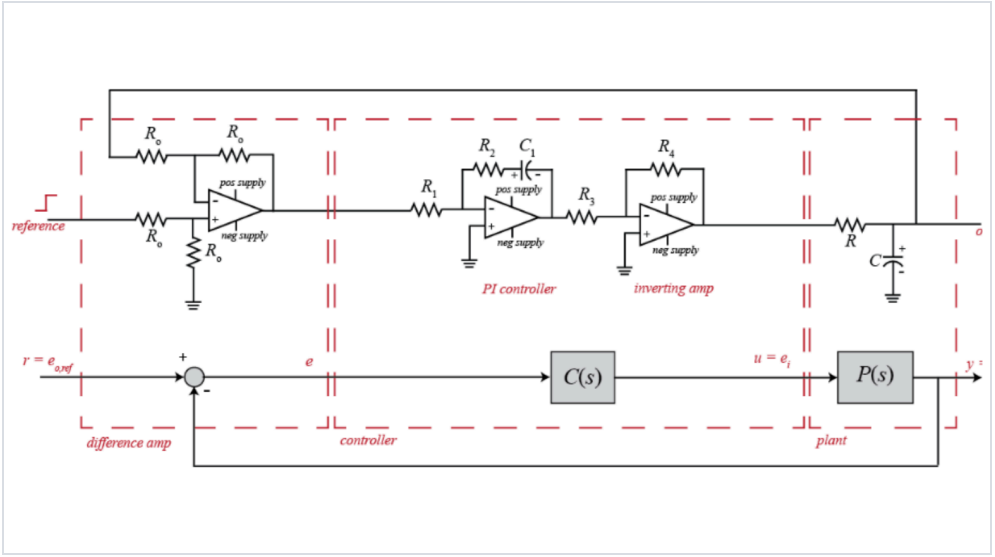
DESIGN

Kp 15 · Ki 145

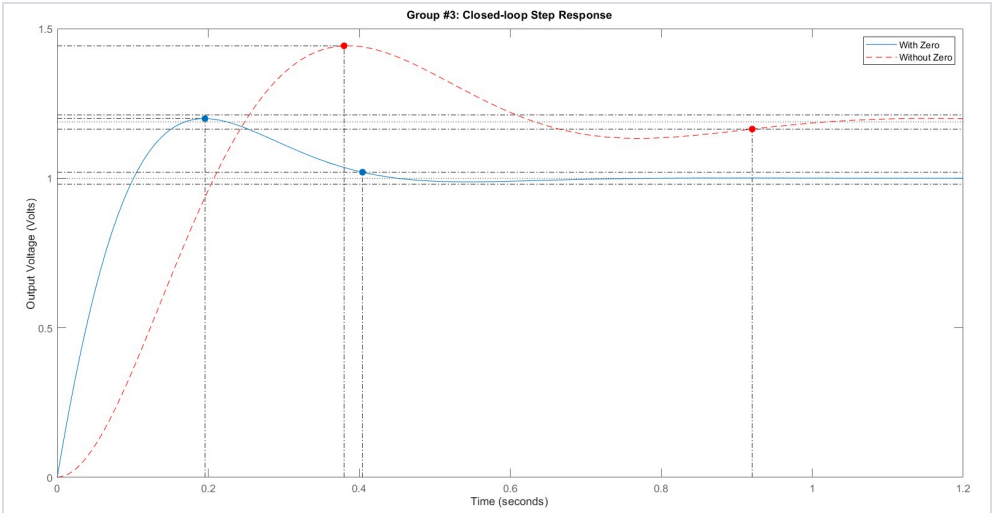
gains built into the op-amp circuit



DESIGN **Pole placement by hand:** timing specs as s-plane boundaries; poles placed inside.



DESIGN **Op-amp loop:** error amp → PI stage → RC plant.



SIMULATION **Predicted step response:** the PI zero speeds the rise but adds ~20% overshoot.

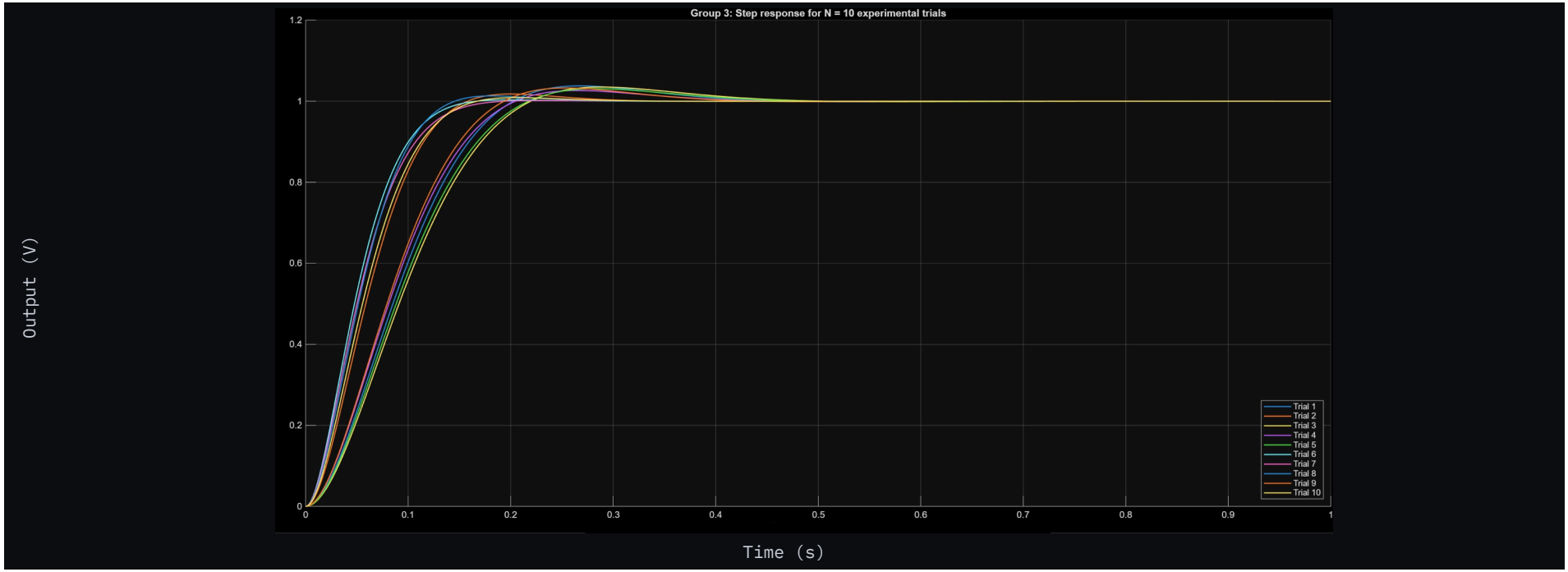
PI control: hardware test

10 hardware trials · One-tailed t-test · TL061 / TL062

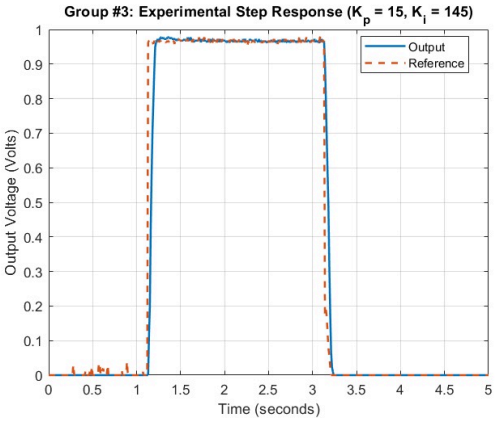
MEASURED · MEAN
0.237 s
peak time (< 0.350 s)

MEASURED · MEAN
0.283 s
2% settling (< 1 s)

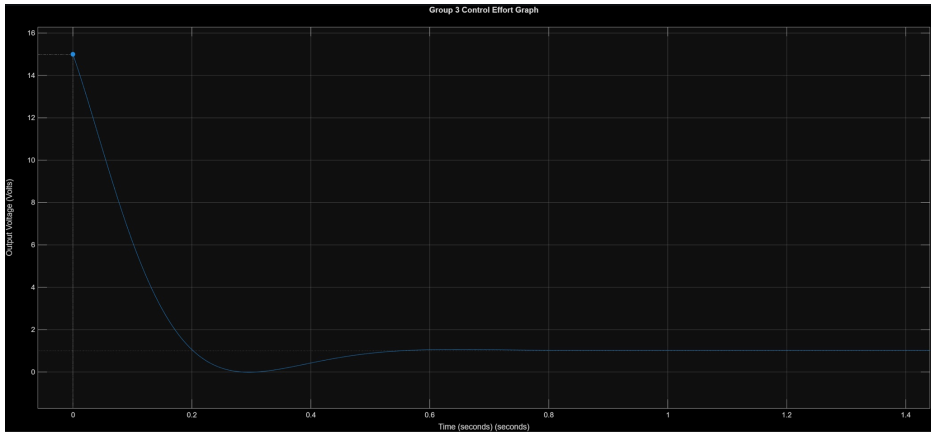
MEASURED · MEAN
2.07%
overshoot (< 25%)



MEASURED **Ten step responses from the op-amp circuit.** Timing and overshoot specs met (t-test, n = 10). Axis titles corrected from the original plot.



MEASURED **Reference tracking** of a 1 V pulse.



SIMULATION **Control effort peaks at 15 V — the 7.3 V effort target was not met.**

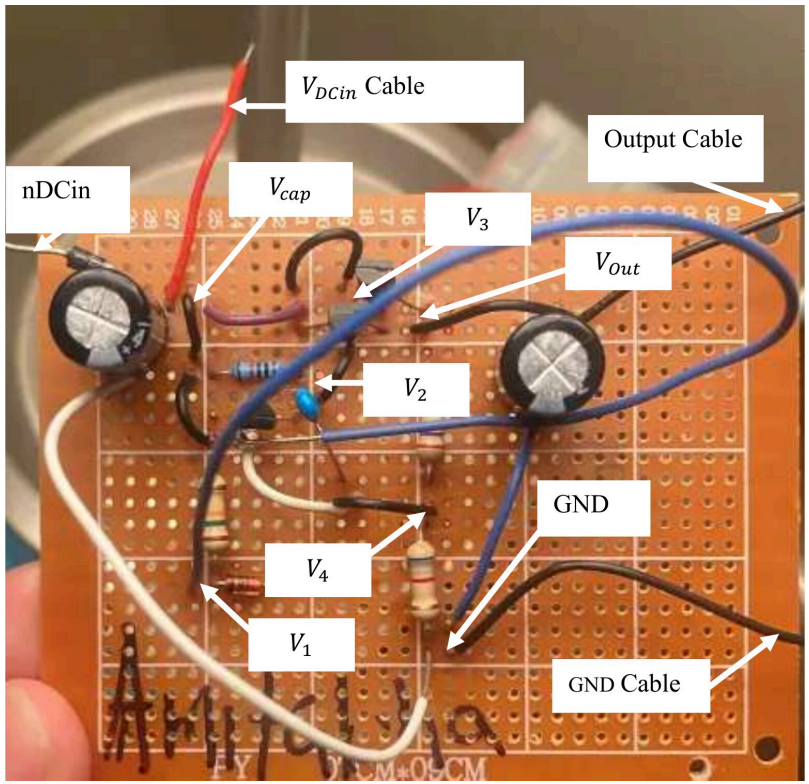
Discrete 7 V linear regulator

BJT + Zener design · LTspice · Soldered prototype

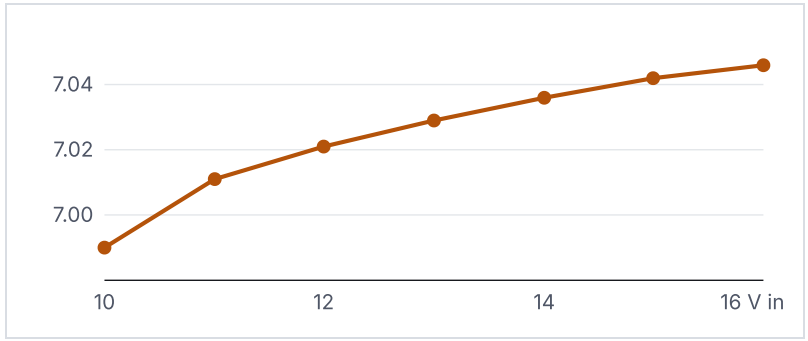
MEASURED

56 mV

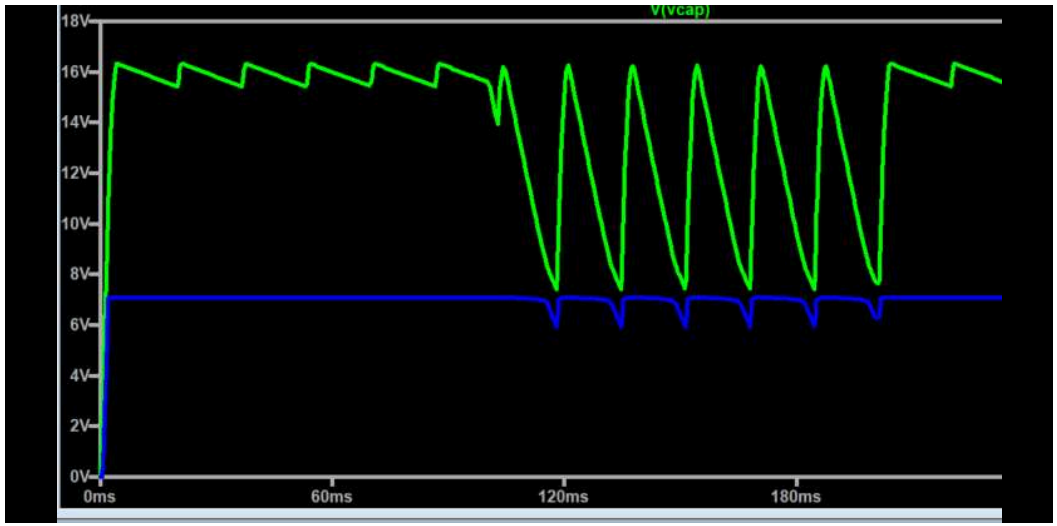
output variation over a 10–16 V input



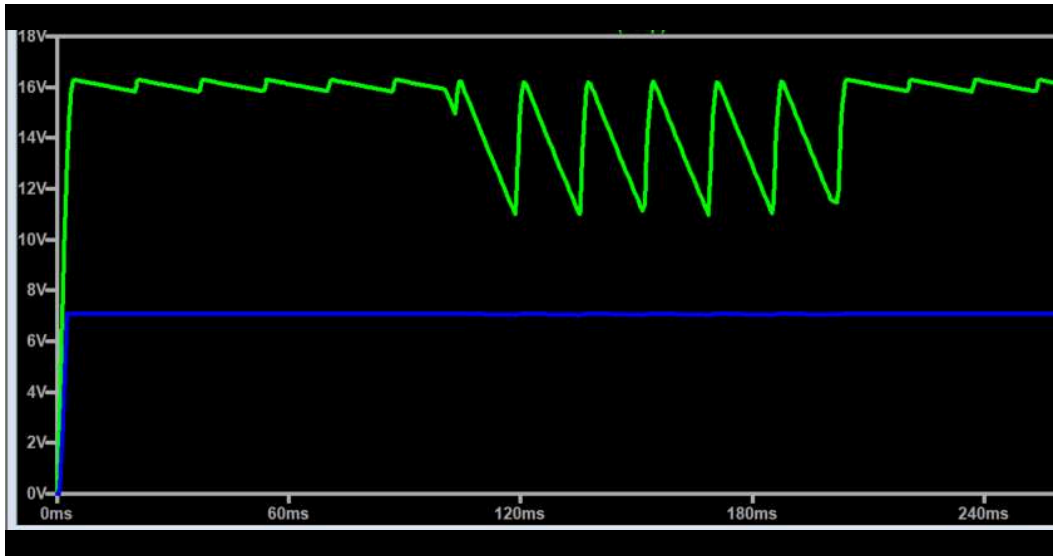
HARDWARE **Soldered prototype,** test nodes labelled to the schematic.



MEASURED **Bench test:** 6.990–7.046 V out, 10–16 V in.



SIMULATION **LTspice, 142 µF:** at 100 mA the capacitor voltage (green) drops below 10 V; output (blue) dips.



SIMULATION **LTspice, 284 µF:** stays above 10 V; output holds flat at ~7 V. Built with 470 µF.

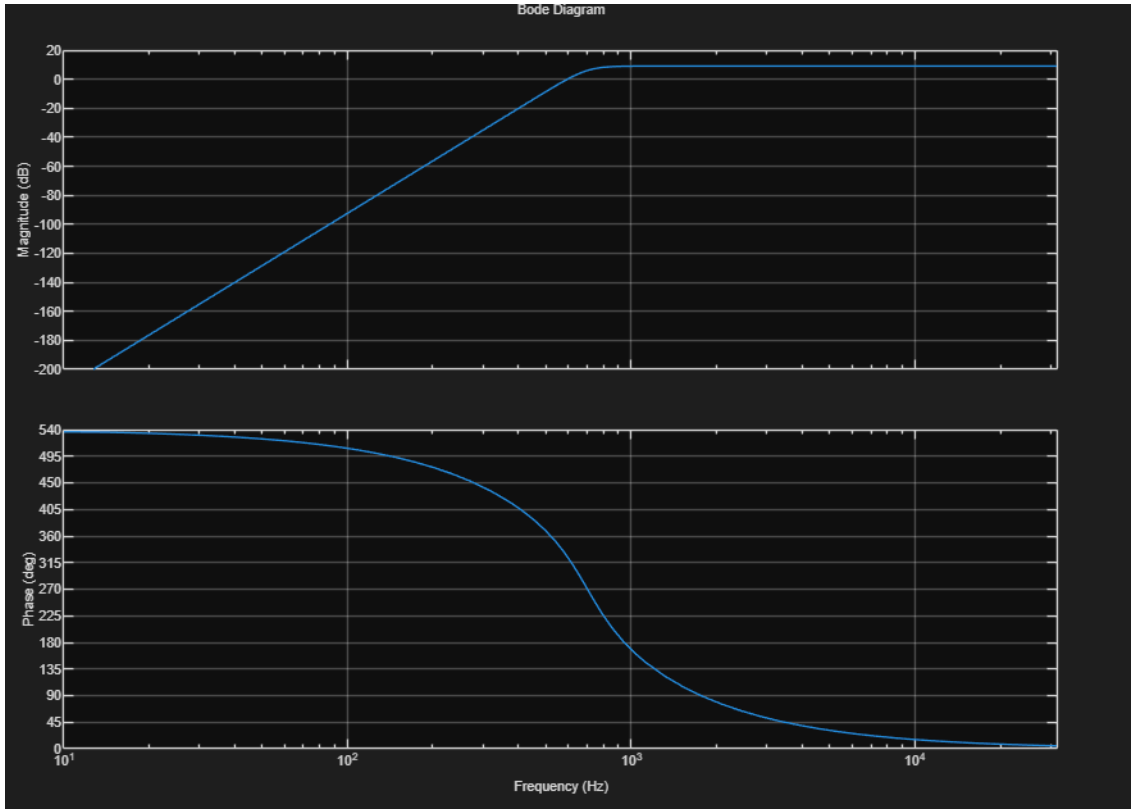
Audio noise-reduction filter

FFT analysis · 6th-order Butterworth HPF · Sallen-Key · Monte Carlo

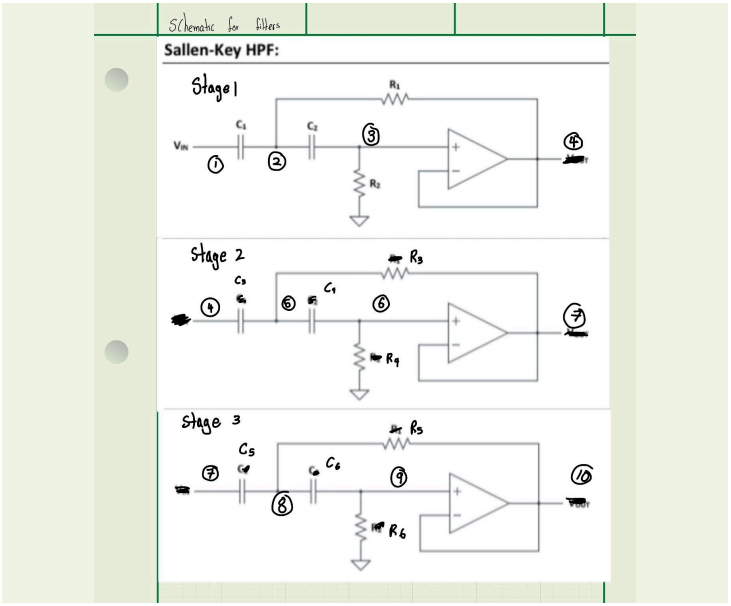
SIMULATED

701.8 Hz

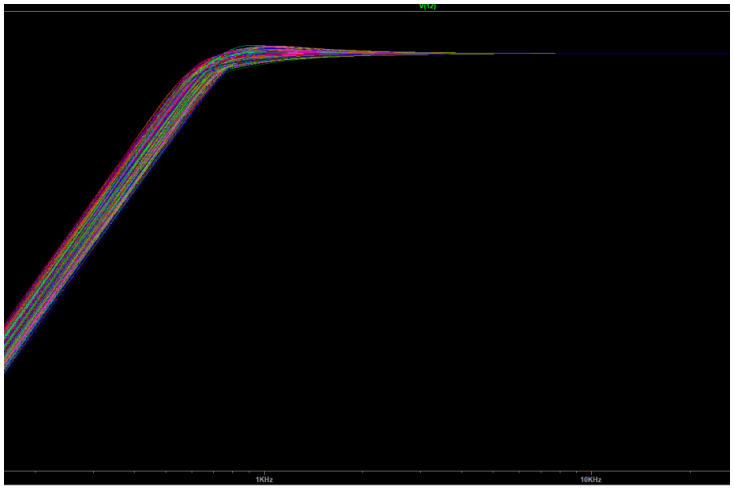
−3 dB cutoff vs 700 Hz spec



SIMULATION **Bode plot (MATLAB):** removes wind (~20–300 Hz), passes bird calls (~1–7 kHz).



DESIGN **Three cascaded Sallen-Key stages** + 9 dB gain stage.



SIMULATION **Monte Carlo, ±20% caps:** cutoff spread → recommended ≤ 10% parts.

EMBEDDED · EECE 344 · SPRING 2026 · INDIVIDUAL

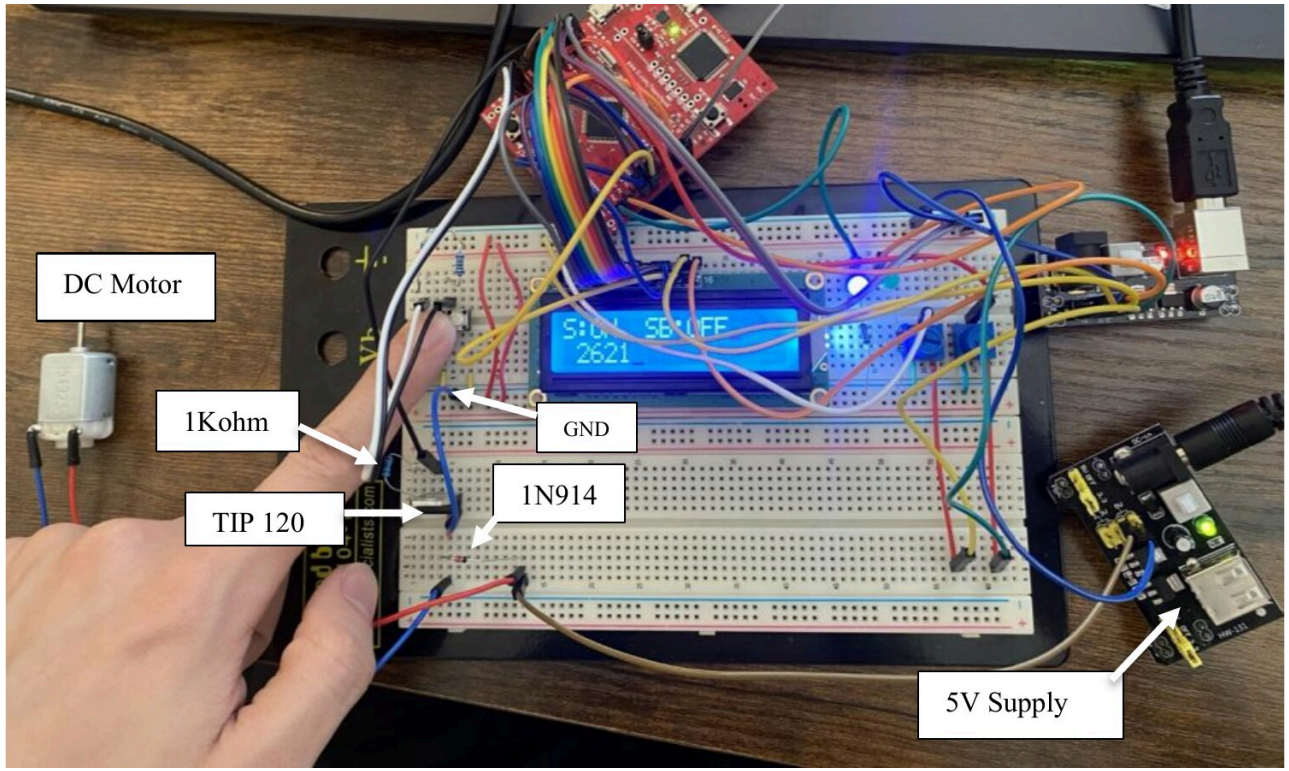
Bare-metal PWM motor control

TM4C123 · Register-level C · Timer-triggered ADC · PWM · TIP120 driver

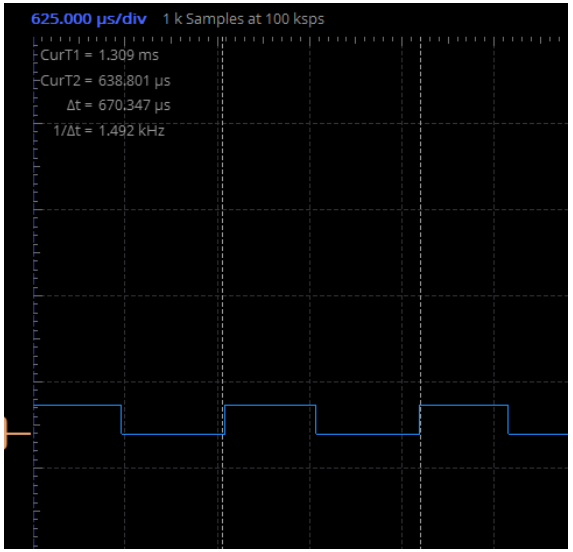
MEASURED

1.492 kHz

PWM vs 1.526 kHz calculated



HARDWARE **Throttle-controlled DC motor:** TM4C123 + LCD, TIP120 low-side driver, 1N914 flyback diode, separate 5 V supply.



MEASURED **PWM on the scope.** Low-throttle twitching fixed by gating PWM below 20% duty.

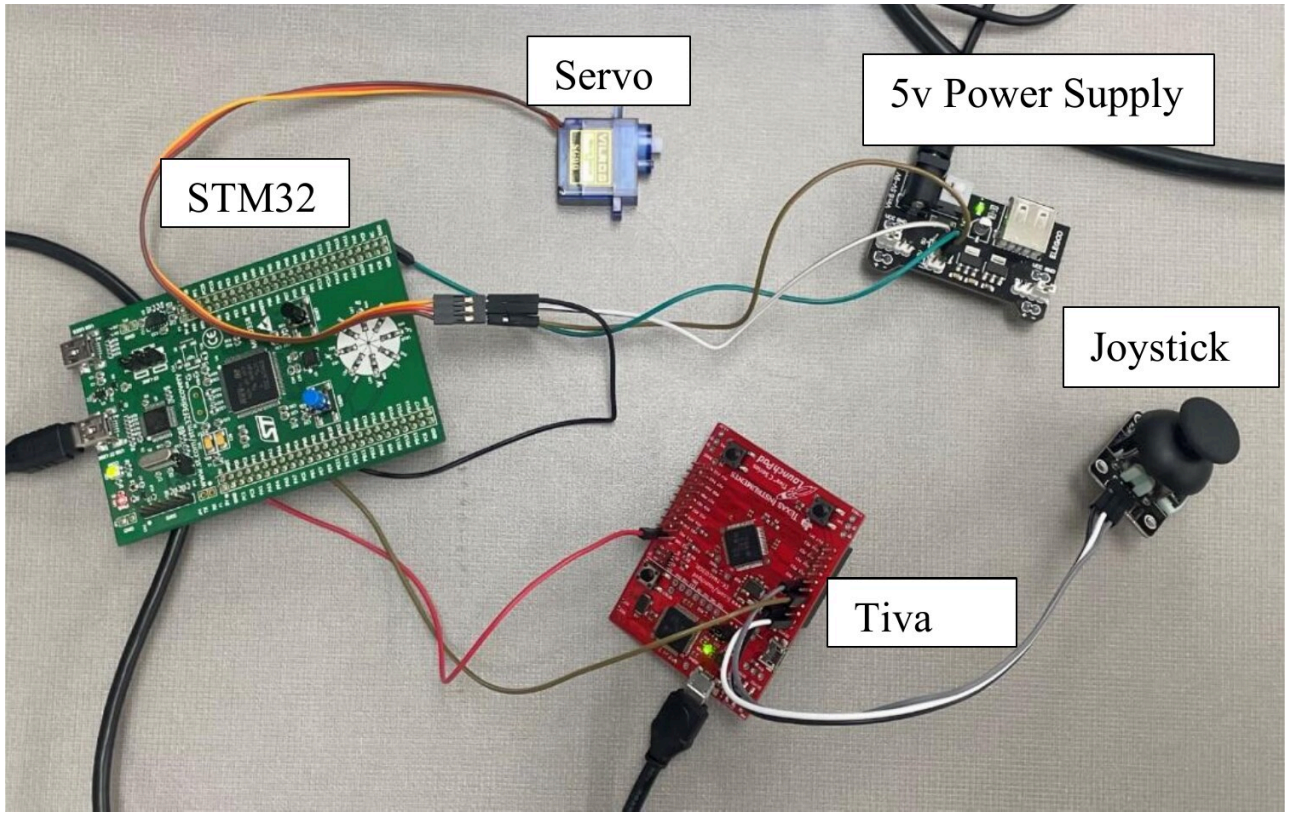
EMBEDDED · EECE 344 FINAL PROJECT · SPRING 2026 · TEAM OF 2

Two-MCU joystick-to-servo

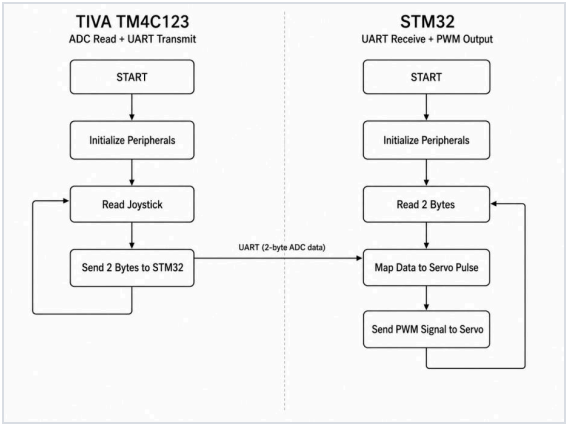
TM4C123 → STM32F303 · UART · Interrupts · ADC · PWM

TESTED
8 µs
deadband that removed servo jitter

CALCULATED
2.08 ms
per 2-byte sample at 9600 baud



HARDWARE **Working system:** joystick → Tiva (ADC) → UART → STM32 (PWM) → servo.



DESIGN **Firmware flow** for both boards.

EMBEDDED · INDEPENDENT PROJECT · SUMMER 2026

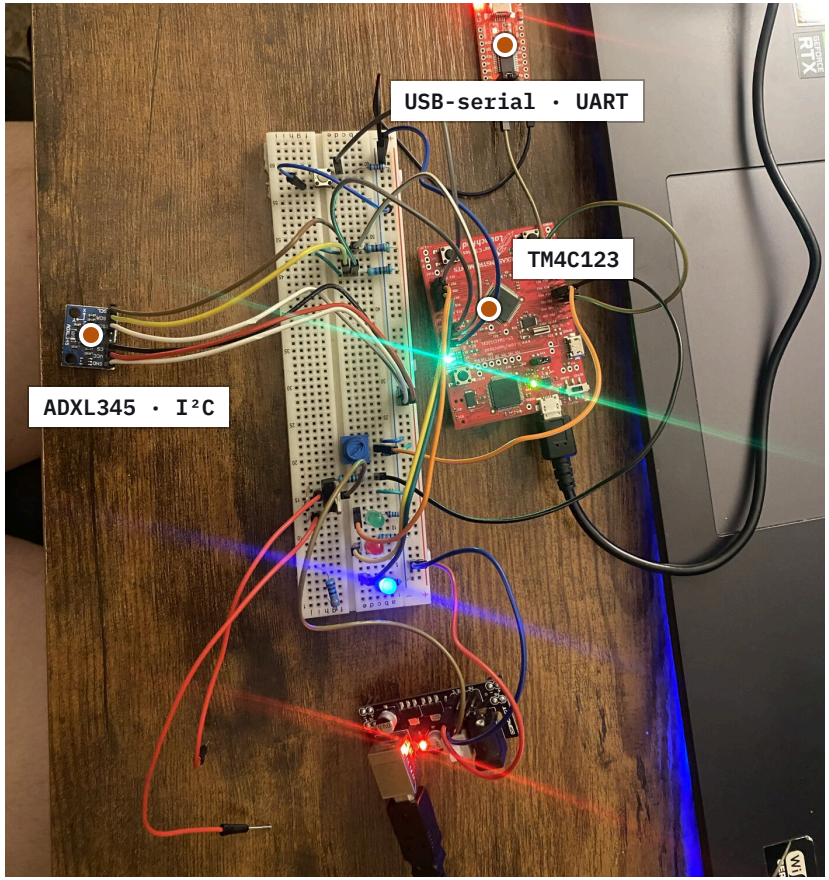
Vibration monitoring & FFT

ADXL345 · Bare-metal I²C · UART 115200 · MATLAB FFT

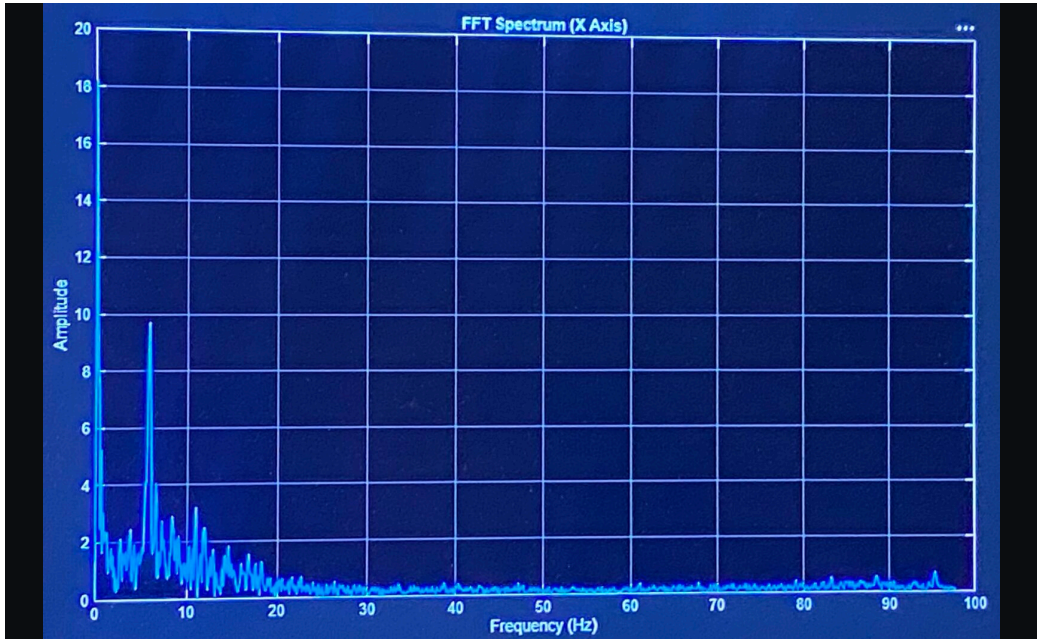
MEASURED

≈ 6 Hz

dominant vibration frequency



HARDWARE **Measurement rig** running my bare-metal drivers.



MEASURED **X-axis spectrum (MATLAB FFT):** dominant component near 6 Hz.



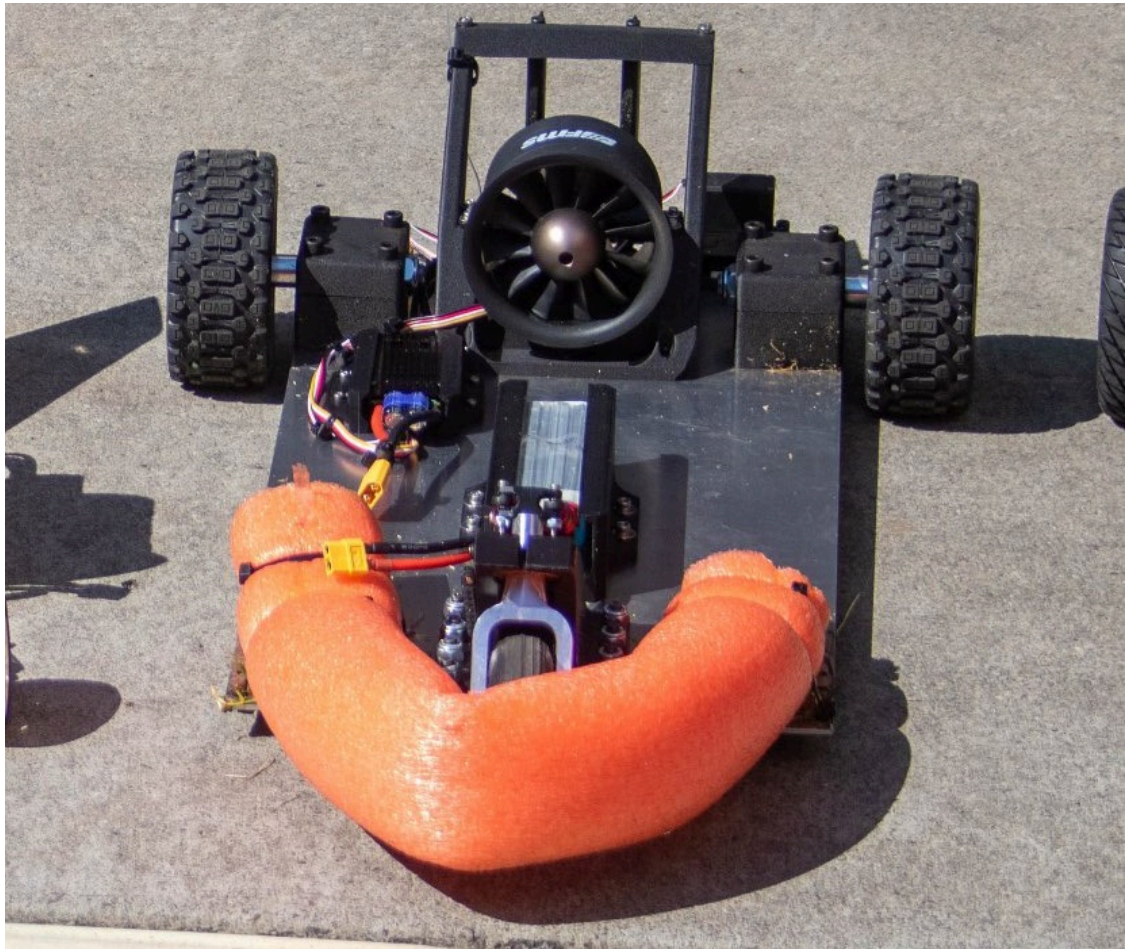
MEASURED **Live 3-axis stream:** rest, disturbance, rest.

MECHANICAL · MECH 340W · SPRING 2026 · TEAM OF 4

Ducted-fan car

Steering & damper design · SolidWorks · Machining · FEA

1ST PLACE · ME DESIGN COMPETITION



HARDWARE **The finished car:** ducted fan and twin rudders at the rear, damped single front wheel.



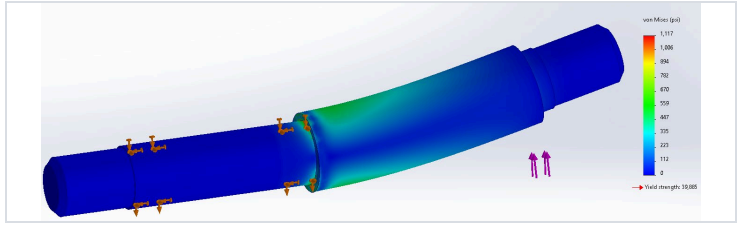
HARDWARE **Competition day** lineup.



DESIGN **My damped front steering.**



HARDWARE **Machined aluminum axle shafts** (my part).



FEA **Rear axle, worst case:** 1117 psi peak, safety factor 35.7.